



**AI Driven SQA: Manual & Automation Testing**

## **Assignment 02**

**Module: 02**

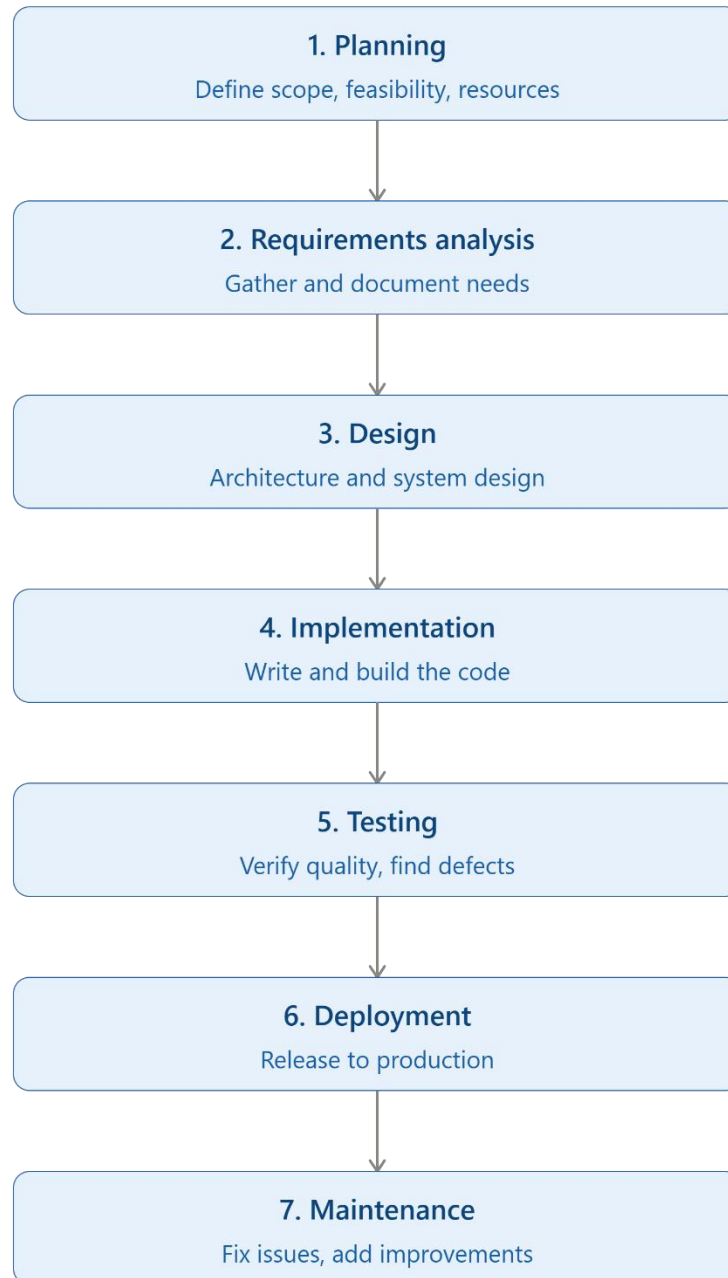
**Topic: Software Development Life Cycle (SDLC)  
in the Age of AI**

**Submitted by,**

**Name: Tahsinur Rahman Tanjim**

**Batch: 21**

Q1. Draw the SDLC lifecycle diagram and briefly explain the workflow.



The SDLC (Software Development Lifecycle) is a structured process teams follow to build software efficiently and predictably. Here's how each phase flows into the next:

1. **Planning** - Determine feasibility, estimate costs and time, and define high-level project scope.
2. **Requirements analysis** – Discuss with interested parties to find out exactly what the software needs to do.

3. **Design** – From requirements to architecture: system architecture, data models, interfaces.
4. **Implementation** – Developers write code based on the design.
5. **Testing** - QA checks that the software works as intended and finds defects before release.
6. **Deployment** – The tested software is released to the production environment for use by users.
7. **Maintenance** – ongoing bug fixes, patches and improvements based on real world usage which often feeds new requirements back into planning, restarting the cycle.

## **Q2. Practical AI Exercise — E-commerce Application**

### **Step 1: Five User Stories for an E-commerce Application**

User Stories -1: As a customer, I want to search products by keyword, so that I can find products I am interested in buying quickly.

User Stories -2: As a customer I want to add products to a shopping cart so I can purchase multiple products in one checkout

User Stories -3: As a customer I want to follow the status of my order so I know when my package will arrive.

User Stories -4: As an admin, I want to manage product inventory, so that I can update stock levels and avoid overselling.

User Stories -5: As a customer I want to be able to save items to a wishlist so that I can buy them later.

### **Step 2: One User Story → Acceptance Criteria**

Selected Story: User Stories -3: As a customer I want to follow the status of my order so I know when my package will arrive.

#### **Acceptance Criteria:**

1. The customer has placed an order and when he opens “My Orders” he can see the current status of his orders (Processing, Shipped, Out for Delivery, Delivered).
2. The order details page displays an estimated delivery date.
3. Customer receives a notification via email/SMS each time the status
4. If a courier tracking number exists for an order, the corresponding order status page displays the tracking number as well as a link to the courier's website to track the package online.
5. Updater from the backend/warehouse system will update

6. If the order is delayed, the system displays a delay notice with a reason, if available.

### Step 3: AI-Generated Sprint Backlog

**Sprint Goal:** Deliver the core shopping experience — search, cart, and order tracking — for the E-commerce. Sprint Duration: 2 weeks , Team Velocity (assumed): 20 points

| ID   | User Story                   | Priority | Story Points | Key Tasks                                                           | Status            |
|------|------------------------------|----------|--------------|---------------------------------------------------------------------|-------------------|
| US-1 | Product search               | High     | 5            | Build search API, add search bar UI, implement keyword indexing     | To Do             |
| US-2 | Add to cart                  | High     | 5            | Cart data model, add/remove/update cart UI, persist cart in session | To Do             |
| US-3 | Order tracking               | High     | 8            | Order status DB field, notification service, tracking UI page       | To Do             |
| US-4 | Inventory management (admin) | Medium   | 5            | Admin dashboard, stock update form, low-stock alert                 | To Do             |
| US-5 | Wishlist                     | Low      | 3            | Wishlist DB table, add/remove UI, wishlist page                     | Backlog (stretch) |

### Step 4: Manual Review and Improvements

#### What the AI did well

1. well-structured and in the correct standard format of user stories created on first try.
2. Covered multiple user roles not just one perspective.
3. The Acceptance Criteria for each feature was specific and testable.
4. The Sprint Backlog logically grouped tasks, assigned priority, and estimated story points, indicating understanding of Agile/Scrum conventions.
5. Saved a lot of time compared to writing all of this from scratch.

#### Corrections made manually

1. US-4 reworded , clarified “manage inventory” to explicitly include “update stock levels and avoid overselling,” the real business value.
2. Adjusted story points increased order tracking (US-3) from 5 to 8 points due to complexity of notifications and courier integration.
3. Added “delay notice” acceptance criterion (6). This common e-commerce edge case was missed by the AI’s first draft.
4. Flagged capacity mismatch moved US-5 (Wishlist) to backlog after total points vs assumed velocity calculation.
5. Rewrote acceptance criteria in Given/When/Then format for easier testability.

## Why human review is still necessary

1. Business context and priorities: AI does not know your real customers or business objectives, it generalizes from shared e-commerce patterns.
2. Capacity and team velocity: AI is unable to tell you your team's particular velocity or technical debt. Estimates need to be sanity checked by humans.
3. Edge cases and domain nuance: AI often only produces the "happy path" and misses exceptions like failed payments or guest checkout.
4. Consistency and quality control: A human makes sure that the output is consistent with the team's standards and existing backlog conventions.
5. Accountability: Backlog decisions must be owned by a human (Product Owner/Scrum Master) the AI is a drafting assistant, not a decision-maker.

## Screenshots of AI Interaction

Generate five user stories for an E-commerce application using the format: As a [user], I want [feature], so that [benefit]

Here are five user stories for an e-commerce application:

1. **As a** first-time visitor, **I want** to browse products without creating an account, **so that** I can explore what's available before committing to sign up.
2. **As a** shopper, **I want** to filter and sort products by price, category, and customer rating, **so that** I can quickly find items that match my needs and budget.
3. **As a** returning customer, **I want** to save items to a wishlist, **so that** I can keep track of products I'm interested in and purchase them later.
4. **As a** buyer, **I want** to view the real-time status of my order (processing, shipped, delivered), **so that** I know when to expect my delivery and can plan accordingly.
5. **As a** customer, **I want** to save multiple shipping addresses and payment methods to my account, **so that** I can check out faster on future purchases.

*PC: User Stories*

Step 2: One User Story → Acceptance Criteria

Selected Story: As a customer I want to follow the status of my order so I know when my package will arrive.

Here are acceptance criteria for that story:

**User story:** As a customer, I want to follow the status of my order so that I know when my package will arrive.

1. After checkout, the customer can view an order status page showing the current stage (e.g., Order Placed → Processing → Shipped → Out for Delivery → Delivered).
2. Each status update is time-stamped, showing when that stage occurred.
3. Once the order ships, the system displays a carrier tracking number and a link to the carrier's tracking page.
4. The customer receives an automatic notification (email and/or SMS, per their preference) whenever the order status changes.
5. The status page displays an estimated delivery date, and updates that estimate if a delay is detected.
6. If tracking data is temporarily unavailable (e.g., carrier API down), the customer sees a clear "status update pending" message rather than a blank page or error.

Write a message...



Sonnet 5 Medium ▾



*PC: Acceptance Criteria*

Using the 5 user stories above, create a Sprint Backlog table for a 2-week sprint. Include columns for ID, User Story, Priority, Story Points, Key Tasks, and Status. Assume a team velocity of about 20 points and flag if the committed points exceed that velocity

Here's the Sprint Backlog based on the 5 user stories:

**Sprint Goal:** Deliver the core shopping experience — search, cart, and order tracking — for the E-commerce MVP.

**Sprint Duration:** 2 weeks | **Team Velocity (assumed):** ~20 points

| ID   | User Story     | Priority | Story Points | Key Tasks                                                           | Status |
|------|----------------|----------|--------------|---------------------------------------------------------------------|--------|
| US-1 | Product search | High     | 5            | Build search API, add search bar UI, implement keyword indexing     | To Do  |
| US-2 | Add to cart    | High     | 5            | Cart data model, add/remove/update cart UI, persist cart in session | To Do  |
| US-3 | Order tracking | High     | 5            | Order status DB field, notification service, tracking UI page       | To Do  |
| US-4 | Inventory      | Medium   | 5            | Admin dashboard, stock update form,                                 | To Do  |

Write a message...

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Sonnet 5 Medium ▾



Claude is AI and can make mistakes. Please double-check responses.

*PC: Sprint Backlog*